VIVEK KUMAR

Data Engineer • PySpark & Apache Kafka • AWS Big Data • Medallion Architecture • GenAI / LangChain

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PROFESSIONAL SUMMARY

Associate Engineer at Virtusa and Results-driven Data Engineer with demonstrated expertise architecting batch and real-time streaming data pipelines, distributed data processing with PySpark, and cloud data lakehouses on AWS. Proficient in Medallion Architecture, Kafka event streaming, and GenAI-powered conversational data analytics (LangChain, Text-to-SQL agents, RAG over data catalogs, and vector databases). Dual AWS Certified (Data Engineer – Associate & Developer – Associate).

TECHNICAL SKILLS

Big Data & Streaming: Apache Spark (PySpark), Apache Kafka, PySpark Structured Streaming, Delta Lake, Airflow, Distributed Computing

Generative AI & LangChain: LangChain (LCEL), LangGraph (Self-Correcting Agents), Text-to-SQL Chains, Vector Search (ChromaDB, pgvector), Semantic Chunking, RAG over Data Dictionaries, OpenAI APIs

Cloud & AWS Data Stack: AWS (S3, Glue, Athena, EMR, Redshift, RDS PostgreSQL, CloudWatch), Databricks, Docker, Terraform

Data Modeling & Arch: Medallion Architecture (Bronze/Silver/Gold), Star/Snowflake Schema, Data Warehousing, Data Lakehouse, Parquet

Languages & DBs: Python, SQL (Advanced Query Optimization, Window Functions), Java, Bash, PostgreSQL, Redis, MongoDB

BI & Analytics: Power BI, Tableau, Metabase, Matplotlib, Data Quality Validation

PROFESSIONAL EXPERIENCE

Associate Engineer | Virtusa (India) Dec 2025 – Present

- Develop and monitor scalable distributed data pipelines and ETL/ELT workflows using PySpark, SQL, and Apache Kafka for enterprise systems.
- Design normalized relational and star-schema analytical models on AWS RDS PostgreSQL and S3, ensuring high throughput and data integrity.
- Construct intelligent LangChain Text-to-SQL query agents allowing stakeholders to query operational data lakes using natural language prompts.
- Implement automated data quality checks, schema evolution validations, and error quarantine pipelines in Python and SQL.

KEY PROJECTS

GenAI Text-to-SQL Data Warehouse Assistant (LangChain & LCEL) | Python, LangChain, LangGraph, LCEL, OpenAI API, PostgreSQL, ChromaDB GitHub

- Architected autonomous conversational data analyst translating natural language business inquiries into safe, executable, read-only SQL queries.
- Implemented dynamic schema-pruning using ChromaDB vector store over table metadata and a LangGraph self-correcting loop that captures SQL execution errors and iteratively re-prompts the LLM for correction.
- Integrated semantic few-shot example selection and guardrails, achieving 92%+ SQL syntax accuracy without exposing sensitive table data.

Real-Time Banking Fraud Detection Streaming Pipeline | Apache Kafka, PySpark Structured Streaming, Delta Lake, Docker, PostgreSQL GitHub

- Constructed streaming fraud detection pipeline ingesting transaction streams via Kafka and processing stateful aggregations with PySpark Structured Streaming.
- Implemented Medallion Architecture: Bronze (raw JSON ingestion to Parquet), Silver (schema validation & deduplication), Gold (real-time velocity scoring).
- Deployed multi-container cluster via Docker Compose, enabling sub-second detection and alert dispatch on high-risk transactions.

Batch Sales ETL & Cloud Data Warehouse Pipeline | PySpark, AWS S3, PostgreSQL, Delta Lake, Power BI, Python GitHub

- Engineered distributed batch ETL pipeline extracting multi-source POS logs and staging clean star-schema datasets into PostgreSQL/AWS S3.
- Automated dirty data handling (duplicate transactions, missing customer keys, negative balances) with quarantine error-logging.
- Configured interactive Power BI reporting dashboards connected to analytical views for real-time sales trend forecasting.

EDUCATION & CERTIFICATIONS

Bachelor of Technology (B.Tech) in Computer Science & Engineering | DIT University, Dehradun 2021 – 2025

Certifications: AWS Certified Developer – Associate (DVA-C02) • AWS Certified Data Engineer – Associate (DEA-C01) • Node.js & Express (Coursera) • React Frontend Development (freeCodeCamp) • Agile Project Management (LinkedIn Learning)